User guide

The BMARIA-12RH is an interregional computable general equilibrium (ICGE) model, implemented using GEMPACK and designed for policy analysis in Brazilian hydrographic regions. RunGEM is a windows program that makes it easy to run any CGE model created with GEMPACK. Customized RunGEM is a particular version of RunGEM that has been hard-wired to work only with one or a few particular models (such as BMARIA-12RH ICGE). This document assumes that a recent version of Customized RunGEM (dated March 2001 or later) that contains the BMARIA-12RH ICGE model is installed.

RunGEM uses a tabbed notebook or card index interface. Model/Data page contains information about files used by the BMARIA-12RH ICGE model. For simulations, the last pages Closure - Shocks - Output files - Solve - Results are usually accessed in that order (from left to right).

Take a look at RunGEM's Model/Data page. It gives two pieces of information:

- The model is RH12.EXE, an executable program. This has been produced by the GEMPACK program TABLO using, as input, the text file RH12.TAB. To change the model specifications, you need to (a) edit RH12.TAB, and (b) run TABLO to make RH12.EXE. That procedure is not covered in this introductory document, and TABLO is not supplied with the Customized RunGEM package.
- There are four input data files. MDATA1.HAR contains the database with interregional input-output and parameters information. PDATA1.HAR contains data about population and labour force. Terminal_LR.HAR is a database used in closure. RH12.TAB is the code file of the BMARIA-12RH model.

In the Closure page, there is a list of the exogenous variables in the currently selected closure. You can choose between several different closures when running a simulation. RunGEM allows you to load different closures already prepared. To see this, use the Load Closure button to load the long-run closure file (RH12.CLS).

Now go to the Shocks page. Click on the Clear Shocks List button to remove whatever shocks are shown. Now you will specify a shock list. Batch files (.bat) containing the shock lists are already provided. Simply double-click the desired .bat file to run all associated simulations automatically. The contents of the batch (.bat) files are summarized in Table 1.

Output controls the names of output files produced by the simulation. Don't change anything now. Go to the Solve page of RunGEM. First, click on the Change button (the one with the Solution method before it). Click on Johansen to select Johansen's method (or Euler/Gragg method). Now click on the Solve button. A "Please Wait" window will appear while the model is solved. Then RunGEM will show you a box telling you how long the solution took. Just press OK. The next natural step is to look at the results.

The Help menu item gives access to extensive online help about RunGEM. Customized RunGEM is a slightly simplified version of RunGEM, so some of the options described there may not apply. There is a Help menu item, "Customized RunGEM Help", dealing with these differences.

Table 1 - Summary of Batch (.bat) Files

SSP2-4.5					SSP5-8.5				
Shock	Climate chenge scenario	Year	Shock name	File name: .bat_variacao	Shock	Climate chenge scenario	Year	Shock name	File name: .bat_variacao
1	ssp245	2015	shock18	1	1	ssp585	2015	shock104	15
2	ssp245	2016	shock19	1	2	ssp585	2016	shock105	15
3	ssp245	2017	shock20	1	3	ssp585	2017	shock106	15
4	ssp245	2018	shock21	1	4	ssp585	2018	shock107	15
5	ssp245	2019	shock22	1	5	ssp585	2019	shock108	15
6	ssp245	2020	shock23	1	6	ssp585	2020	shock109	15
1	ssp245	2021	shock24	2	1	ssp585	2021	shock110	16
2	ssp245	2022	shock25	2	2	ssp585	2022	shock111	16
3	ssp245	2023	shock26	2	3	ssp585	2023	shock112	16
4	ssp245	2024	shock27	2	4	ssp585	2024	shock113	16
5	ssp245	2025	shock28	2	5	ssp585	2025	shock114	16
6	ssp245	2026	shock29	2	6	ssp585	2026	shock115	16
1	ssp245	2027	shock30	3	1	ssp585	2027	shock116	17
2	ssp245	2028	shock31	3	2	ssp585	2028	shock117	17
3	ssp245	2029	shock32	3	3	ssp585	2029	shock118	17

4	ssp245	2030	shock33	3	4	ssp585	2030	shock119	17
5	ssp245	2031	shock34	3	5	ssp585	2031	shock120	17
6	ssp245	2032	shock35	3	6	ssp585	2032	shock121	17
1	ssp245	2033	shock36	4	1	ssp585	2033	shock122	18
2	ssp245	2034	shock37	4	2	ssp585	2034	shock123	18
3	ssp245	2035	shock38	4	3	ssp585	2035	shock124	18
4	ssp245	2036	shock39	4	4	ssp585	2036	shock125	18
5	ssp245	2037	shock40	4	5	ssp585	2037	shock126	18
6	ssp245	2038	shock41	4	6	ssp585	2038	shock127	18
1	ssp245	2039	shock42	5	1	ssp585	2039	shock128	19
2	ssp245	2040	shock43	5	2	ssp585	2040	shock129	19
3	ssp245	2041	shock44	5	3	ssp585	2041	shock130	19
4	ssp245	2042	shock45	5	4	ssp585	2042	shock131	19
5	ssp245	2043	shock46	5	5	ssp585	2043	shock132	19
6	ssp245	2044	shock47	5	6	ssp585	2044	shock133	19
1	ssp245	2045	shock48	6	1	ssp585	2045	shock134	20
2	ssp245	2046	shock49	6	2	ssp585	2046	shock135	20
3	ssp245	2047	shock50	6	3	ssp585	2047	shock136	20
4	ssp245	2048	shock51	6	4	ssp585	2048	shock137	20
5	ssp245	2049	shock52	6	5	ssp585	2049	shock138	20
6	ssp245	2050	shock53	6	6	ssp585	2050	shock139	20
1	ssp245	2051	shock54	7	1	ssp585	2051	shock140	21
2	ssp245	2052	shock55	7	2	ssp585	2052	shock141	21
3	ssp245	2053	shock56	7	3	ssp585	2053	shock142	21
4	ssp245	2054	shock57	7	4	ssp585	2054	shock143	21
5	ssp245	2055	shock58	7	5	ssp585	2055	shock144	21
6	ssp245	2056	shock59	7	6	ssp585	2056	shock145	21
1	ssp245	2057	shock60	8	1	ssp585	2057	shock146	22
2	ssp245	2058	shock61	8	2	ssp585	2058	shock147	22
3	ssp245	2059	shock62	8	3	ssp585	2059	shock148	22
4	ssp245	2060	shock63	8	4	ssp585	2060	shock149	22
5	ssp245	2061	shock64	8	5	ssp585	2061	shock150	22
6	ssp245	2062	shock65	8	6	ssp585	2062	shock151	22
1	ssp245	2063	shock66	9	1	ssp585	2063	shock152	23
2	ssp245	2064	shock67	9	2	ssp585	2064	shock153	23
3	ssp245	2065	shock68	9	3	ssp585	2065	shock154	23
4	ssp245	2066	shock69	9	4	ssp585	2066	shock155	23
5	ssp245	2067	shock70	9	5	ssp585	2067	shock156	23
6	ssp245	2068	shock71	9	6	ssp585	2068	shock157	23
1	ssp245	2069	shock72	10	1	ssp585	2069	shock158	24
2	ssp245	2070	shock73	10	2	ssp585	2070	shock159	24
3	ssp245	2071	shock74	10	3	ssp585	2071	shock160	24
4	ssp245	2072	shock75	10	4	ssp585	2072	shock161	24
5	ssp245	2073	shock76	10	5	ssp585	2073	shock162	24
6	ssp245	2074	shock77	10	6	ssp585	2074	shock163	24
1	ssp245	2075	shock78	11	1	ssp585	2075	shock164	25
2	ssp245	2076	shock79	11	2	ssp585	2076	shock165	25

3	ssp245	2077	shock80	11	3	ssp585	2077	shock166	25
4	ssp245	2078	shock81	11	4	ssp585	2078	shock167	25
5	ssp245	2079	shock82	11	5	ssp585	2079	shock168	25
6	ssp245	2080	shock83	11	6	ssp585	2080	shock169	25
1	ssp245	2081	shock84	12	1	ssp585	2081	shock170	26
2	ssp245	2082	shock85	12	2	ssp585	2082	shock171	26
3	ssp245	2083	shock86	12	3	ssp585	2083	shock172	26
4	ssp245	2084	shock87	12	4	ssp585	2084	shock173	26
5	ssp245	2085	shock88	12	5	ssp585	2085	shock174	26
6	ssp245	2086	shock89	12	6	ssp585	2086	shock175	26
1	ssp245	2087	shock90	13	1	ssp585	2087	shock176	27
2	ssp245	2088	shock91	13	2	ssp585	2088	shock177	27
3	ssp245	2089	shock92	13	3	ssp585	2089	shock178	27
4	ssp245	2090	shock93	13	4	ssp585	2090	shock179	27
5	ssp245	2091	shock94	13	5	ssp585	2091	shock180	27
6	ssp245	2092	shock95	13	6	ssp585	2092	shock181	27
1	ssp245	2093	shock96	14	1	ssp585	2093	shock182	28
2	ssp245	2094	shock97	14	2	ssp585	2094	shock183	28
3	ssp245	2095	shock98	14	3	ssp585	2095	shock184	28
4	ssp245	2096	shock99	14	4	ssp585	2096	shock185	28
5	ssp245	2097	shock100	14	5	ssp585	2097	shock186	28
6	ssp245	2098	shock101	14	6	ssp585	2098	shock187	28
7	ssp245	2099	shock102	14	7	ssp585	2099	shock188	28
8	ssp245	2100	shock103	14	8	ssp585	2100	shock189	28

Source: authors' elaboration

The detailed results for all simulated scenarios are reported in the file [Detailed results.xlsx].